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ROCKWOOL Rockfit Mono



ISSUED 26.10.2023
VALID UNTIL 26.10.2028

THIRD PARTY VERIFIED
in accordance with EN 15804+A2
and B-EPD-PCR
EN 16783, Thermal insulation products – Product category
rules (PCR) for factory made and in-situ formed products for
environmental product declarations (2017)

MODULES DECLARED

1 m² Rockfit Mono insulation for cavity walls, curtain walls
and separating walls.

A123	A4	A5	B2 B4	C	D
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The intended use of this EPD is to communicate scientifically based environmental information for construction products, for the purpose of assessing the environmental performance of buildings. This EPD is only valid when registered on www.b-epd.be. The FPS Public Health cannot be held responsible for the information provided by the owner of the EPD.

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1 PRODUCT DESCRIPTION

1.1 Product name

Rockfit Mono, mineral wool, 600x1000 mm, 35 kg/m³,
 λ 0,035W/m.K

1.2 Product description and intended use

Mineral wool insulation. The finished product is included in this EPD, but not the materials needed for installation.

Declared for 150 mm. Available in 40 to 250 mm thickness.

The product is a rigid slab and has no facings.

Rockfit Mono is a flexible and water-repellent cavity board. Suitable for thermal and acoustic insulation of completely or partially filled cavity walls.

In the thinner thicknesses, Rockfit Mono is also extremely suitable as thermal and acoustic break in house partition walls (common walls).

This EPD is a specific EPD that reflects results from a single company and represents the manufacturing for product available on the Belgian market.

This product is used in cavity walls, curtain walls and separating walls.



1.3 Reference flow / declared unit

1m² of installed insulation product with a defined R_D-Value functioning for a period of 75 years in a building.

Packaging is included.

The weight per reference flow is 5.25 kg.

The density of the product is 35 kg/m³.

The thickness of the product is 150 mm.



1.4 Installation

Materials for fixation and installation are not included. This means this EPD is “as produced”.

Regarding installation this EPD only includes the environmental impact related to the product itself.

A typical installation is manually done with the help of cutting tools and results in 2% waste and is in accordance with EN16783.

Installation requires the need of additional products and materials for which the impact is not included in this EPD and which shall be taken into account at building level.

For auxiliary materials and installation guidelines: see technical data and installation guidelines on www.rockwool.com.



1.5 Composition and content

Components	Composition / content / ingredients	Quantity
Product	– Stonewool	96.8%
	– Resin	3.2%
	– Facing	No facing
Packaging	– Pallets	0.002 kg
	– Foil	0.005 kg

The product does not contain materials listed in the “Candidate list of Substances of Very High Concern for authorization”.

1.6 Reference service life

The reference service life is estimated at 75 years.

The thermal performance characteristics of thermal insulation products are usually based on a minimum of 50 years and stone wool keeps its characteristics even longer. Therefore, a RSL for a ROCKWOOL insulation product is recommended to be at least 50 years or according to the lifetime of the construction that it is a part of. This is supported also by the PCR for thermal insulation products¹.

The stable thermal characteristics of mineral wool insulation products and their long-term durability are supported by the durability study run by EURIMA in 2016². Based on this information and product use in buildings in the Benelux, ROCKWOOL considers the reference service life of ROCKWOOL products to be the service life for residential buildings and is set to 75 years. This is a scenario as it relates to the service life of the application and the building.

The conditions under which this RSL is valid are as following: installation to be conducted in accordance with manufacturers guidelines. It is assumed that the manufacturer’s instructions are clear and followed. Any uncertainty, and manufacturer should be contacted for instruction.

1.7 Description of geographical representativity

The EPD is representative for the Belgian market.

¹ EN 16783:2014: PCR for thermal Insulation Products- 6.3.3 Reference Service Life RSL

² FIW Munchen e.V. (January 2016): Durability Project Mineral Wool for EURIMA Association



1.8 Description of the production process and technology

Stone and briquettes are melted in a cupola and melted material is spun on wheels that create stone fibres. These fibres are collected into a mat while resin is applied according to production specification. The resin is cured, and mineral oil is applied. The mat moves through a line process where compression, cutting and finishing are done to create the insulation products. ROCKWOOL products are then packaged. Mostly on pallet and a plastic wrap. Internal waste is recycled on-site.

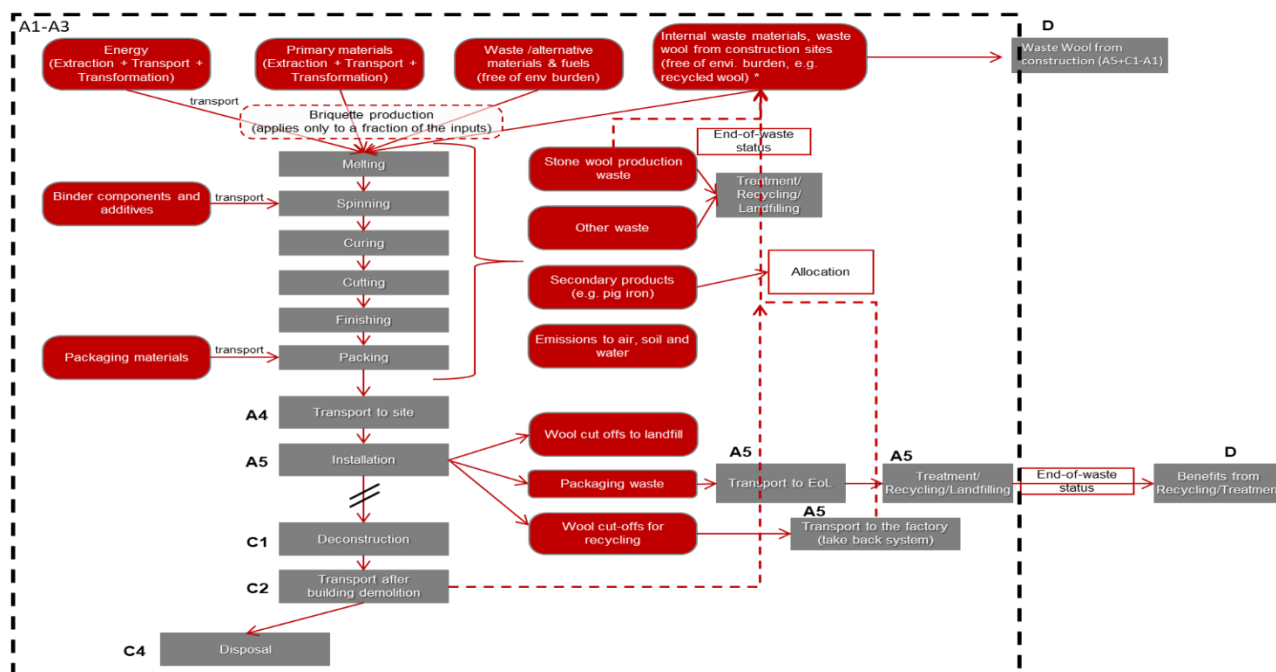
All raw materials that enter the manufacturing stage have been traced back to the point of extraction from nature. In addition to raw materials, secondary materials from the market and internally recycled materials are used. Internally recycled material is modelled without environmental burden, apart from the processing it takes to prepare for on-site recycling.

Blast Furnace Slag and the Converter slag are by-products of the steel industry and therefore the related emissions, attributed to their production should be accounted for. For this purpose, the principles of economic allocation, between the production of steel and slag are used. This B-EPD uses a conservative value of 1% using the process for pig iron.

The raw material input aluminium oxide (alumina source) is also classified as a by-product from the aluminium production and the allocation assigned to that is also 1% following a conservative approach. This B-EPD uses 1% of the process for aluminium, cast alloy.

The system boundaries include all materials, products and energy, as well as waste processing up to the end-of-waste state, a cut-off or disposal of final residues during the product stage. There are no co-products that have a value of 1% or higher and hence only mineral wool products and wastes leave the facility. No offsetting of CO₂ with certificates has been applied.

Information module D aims at transparency for the environmental benefits or loads resulting from reusable products, recyclable materials and/or useful energy carriers leaving a product system, e.g. as secondary materials or fuels. This only applies to packaging waste from module [A5] that is recycled or incinerated.



2 TECHNICAL DATA / PHYSICAL CHARACTERISTICS

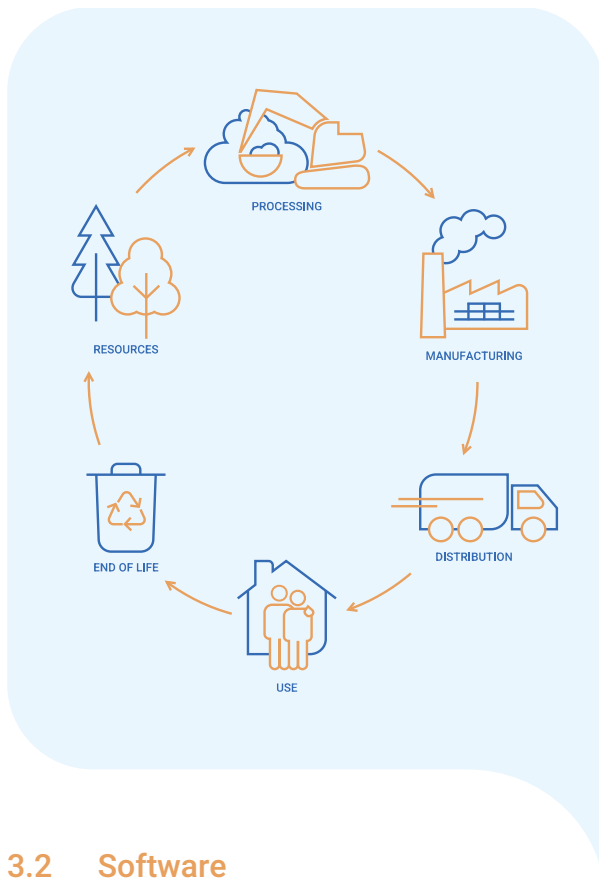
Technical property	Standard	Value	Unit	Comment
Thickness		150	mm	
Thermal characteristics (λ_D)	EN 13162	0.035	W/m.K	



3 LCA-STUDY

3.1 Date of LCA-study

October 2023



3.2 Software

For the calculation of the LCA results, the software program SimaPro 9.1 has been used.

3.3 Information on allocation

Co-product allocation

Co-product is relevant for internal allocation of consumables to products. This is relevant for the stone wool manufacturing data where multiple products are produced on the same production lines. The approach for co-products within the ROCKWOOL facilities was followed using mass and binder content.

Plant level data

Plant level data for ROCKWOOL is limited to the waste produced. Waste has been assigned to product based on the amount of wool in the product. All other data is production line specific and relevant allocation procedures per line are discussed under "co-product allocation" above.

Use of secondary energy and material resources

Some secondary energy and material resources are used at ROCKWOOL for the wool lines. Examples are the use of blast furnace slag and anode cokes. The processes to be able to utilize these secondary resources as well as the transportation to ROCKWOOL are included in the system boundaries. The production of these materials is considered where co-products generate 1% or more of total revenue. An example is blast furnace slag.

Multi-input processes

Various products are processed together within an individual process, e.g. in a waste incineration plant or a landfill site. Allocation is performed based on physical classification of the material flows.

Recycling

Internally produced waste is recycled using the on-site recycling factory that produces briquettes that are used to make new products. The EPD includes this recycling by accounting for the consumption of briquettes on the lines.

Recycling benefits from A5 are reported under module D.

3.4 Informatie on cut off

The cut-off criteria of EN 15804 are met.



3.5 Information on excluded processes

Following processes were excluded for the inventory: flows related to paper product labels, human activities such as employee transport and administration activity.

3.6 Information on biogenic carbon modelling

The product and packaging do not contain biogenic carbon, but the packaging does.

Biogenic carbon content	(kg C / FU)
Biogenic carbon content in product (at the gate)	0
Biogenic carbon content in accompanying packaging (at the gate)	0.17

3.7 Information on carbon offsetting

Carbon offsetting is not allowed in the EN 15804 and hence not taken into account in the calculations.

3.8 Additional or deviating characterisation factors

EN 15804+A2 is followed, meaning the characterization factors from EC-JRC were applied. No additional or deviating characterisation factors were used.

3.9 Description of the variability

The manufacturing facility in Roermond uses multiple production lines and products can be produced on more than one line. Manufacturing data have been averaged per wool line and allocated to the amount of wool for the products and the ratio of the production volume of the specific products per line. The lines are of different age, use different efficiencies, allow for different uses of recycled briquettes and produce different types of wool. Averaging by volume over the lines provides the best way to model these differences into a representative average per product.

3.10 Specificity

The data used for the LCA are specific for this product which is manufactured by a single manufacturer in a single production site.

3.11 Period of data collection

Manufacturer specific data have been collected for the year 2021.

3.12 Information on data collection

Actual consumption data per manufacturing line have been compiled. The most notable background processes for which generic datasets were used are:

- Diesel, low-sulfur {RER}| market group for | Cut-off, U
- Natural gas, high pressure {NL}| market for | Cut-off, U
- 'Heat, district or industrial, natural gas {Europe without Switzerland}| heat production, natural gas, at industrial furnace >100kW | Cut-off, U'
- Electricity, low voltage {NL}| market for | Cut-off, U
- Transport, freight, lorry, unspecified {GLO}| market for | Cut-off
- Transport, freight, inland waterways, barge {RER}| processing | Cut-off, U
- Transport, freight, sea, transoceanic ship {GLO}| market for | Cut-off, U
- 'Electricity, high voltage {NL}| heat and power co-generation, wood chips, 6667 kW, state-of-the-art 2014 | Cut-off, U'
- 'Heat, district or industrial, other than natural gas {NL}| heat and power co-generation, wood chips, 6667 kW, state-of-the-art 2014 | Cut-off, U'
- For sorting: Electricity, low voltage {NL}| market for | Cut-off, U

3.13 Database used for background data

Ecoinvent 3.8

3.14 Energy mix

National energy mix for generic processes. ROCKWOOL has a specific contract from renewable resources for the manufacturing site for the declared product. The following datasets have been used to model the actual supply mix: Electricity, high voltage {EE}| electricity production, wind, >3MW turbine, onshore | Cut-off, U, Electricity, hydropower, at run-of-river power plant/RER U and Electricity, high voltage {IT}| heat and power co-generation, wood chips, 6667 kW, state-of-the-art 2014 | Cut-off, U.



4 PRODUCTION SITES

The declared product is produced in Roermond, The Netherlands.



5 SYSTEM BOUNDARIES

Product stage			Construction installation stage		Use stage							End of life stage				Beyond the system boundaries
Raw materials	Transport	Manufacturing	Transport	Construction installation stage	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☐	☐	☒	☒	☒	☒	☒

X = included in the EPD

☐ = module not declared



6 POTENTIAL ENVIRONMENTAL IMPACTS PER REFERENCE FLOW

	Production			Construction process stage		Use stage							End-of-life stage				
		A2 Transport A2 Transport A3 manufacturing		A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	D Reuse, recovery, recycling
 GWP total (kg CO2 equiv/FU)		4.12E+00		3.80E-01	7.48E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	8.16E-02	0.00E+00	2.90E-02	1.18E-01
 GWP fossil (kg CO2 equiv/FU)		4.88E+00		3.80E-01	1.21E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	8.14E-02	0.00E+00	2.90E-02	-3.77E-02
 GWP biogenic (kg CO2 equiv/FU)		-7.67E-01		1.89E-04	6.19E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	1.87E-04	0.00E+00	4.45E-05	1.55E-01
 GWP luluc (kg CO2 equiv/FU)		1.55E-03		1.65E-04	4.81E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	3.88E-05	0.00E+00	8.70E-06	-6.07E-05
 ODP (kg CFC 11 equiv/FU)		2.25E-07		8.45E-08	7.53E-09	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	1.73E-09	0.00E+00	9.38E-09	-4.92E-09
 AP (mol H+ eq/FU)		4.93E-02		1.86E-03	1.06E-03	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	2.60E-04	0.00E+00	2.12E-04	-1.13E-04
 EP - freshwater (kg P- equiv/FU)		1.81E-04		3.49E-06	3.93E-06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	6.48E-07	0.00E+00	2.63E-07	-9.62E-07
 EP - marine (kg N- equiv/FU)		4.80E-03		6.11E-04	1.23E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	2.60E-04	0.00E+00	7.73E-05	-2.93E-05
 EP - terrestrial (mol N- equiv/FU)		1.36E-01		6.75E-03	2.98E-03	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	9.44E-04	0.00E+00	8.52E-04	-3.24E-04
 POCP (kg Ethene equiv/FU)		1.48E-02		1.94E-03	3.79E-04	7.37E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	3.96E-04	0.00E+00	2.44E-04	-1.03E-04
 ADP Elements (kg Sb equiv/FU)		1.83E-05		1.37E-05	7.74E-07	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	2.44E-07	0.00E+00	5.48E-07	-1.71E-07
 ADP fossil fuels (MJ/FU)		4.81E+01		5.70E+00	1.23E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	1.14E+00	0.00E+00	6.30E-01	-7.71E-01
 WDP (m³ water eq deprived /FU)		5.82E-01		1.75E-02	2.44E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	4.91E-03	0.00E+00	2.05E-03	-1.15E+00

GWP TOTAL = TOTAL GLOBAL WARMING POTENTIAL (CLIMATE CHANGE); GWP-LULUC = GLOBAL WARMING POTENTIAL (CLIMATE CHANGE) LAND USE AND LAND USE CHANGE; ODP = OZONE DEPLETION POTENTIAL; AP = ACIDIFICATION POTENTIAL FOR SOIL AND WATER; EP = EUTROPHICATION POTENTIAL; POCP = PHOTOCHEMICAL OZONE CREATION; ADPE = ABIOTIC DEPLETION POTENTIAL – ELEMENTS; ADPF = ABIOTIC DEPLETION POTENTIAL – FOSSIL FUELS; WDP = WATER USE (WATER (USER) DEPRIVATION POTENTIAL, DEPRIVATION-WEIGHTED WATER CONSUMPTION)

7 RESOURCE USE

	Production			Construction process stage		Use stage							End-of-life stage				
		A2 Transport A2 Transport A3 manufacturing		A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	D Reuse, recovery, recycling
PERE (MJ/FU, net calorific value)		2.46E+01		6.55E-02	1.72E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	1.38E-02	0.00E+00	5.96E-03	-6.93E-03
PERM (MJ/FU, net calorific value)		5.30E+00		3.21E-02	-9.45E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	4.26E-03	0.00E+00	2.70E-03	-1.76E+00
PERT (MJ/FU, net calorific value)		2.99E+01		9.76E-02	1.62E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	1.81E-02	0.00E+00	8.66E-03	-1.77E+00
PENRE (MJ/FU, net calorific value)		7.11E+01		5.72E+00	1.88E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	1.15E+00	0.00E+00	6.06E-01	-8.35E-01
PENRM (MJ/FU, net calorific value)		2.22E-01		0.00E+00	-1.83E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT (MJ/FU, net calorific value)		7.13E+01		5.72E+00	1.70E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	1.15E+00	0.00E+00	6.06E-01	-8.35E-01
SM (kg/FU)		3.56E+00		0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF (MJ/FU, net calorific value)		0.00E+00		0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF (MJ/FU, net calorific value)		4.32E-01		0.00E+00	8.65E-03	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW (m³ water eq/FU)		0.00E+00		0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

PERE = USE OF RENEWABLE PRIMARY ENERGY EXCLUDING RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PERM = USE OF RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PERT = TOTAL USE OF RENEWABLE PRIMARY ENERGY RESOURCES; PENRE = USE OF NON-RENEWABLE PRIMARY ENERGY EXCLUDING NON-RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PENRM = USE OF NON-RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PENRT = TOTAL USE OF NON-RENEWABLE PRIMARY ENERGY RESOURCES; SM = USE OF SECONDARY MATERIAL; RSF = USE OF RENEWABLE SECONDARY FUELS; NRSF = USE OF NON-RENEWABLE SECONDARY FUELS; FW = NET USE OF FRESH WATER

8 WASTE CATEGORIES & OUTPUT FLOWS

	Production			Construction process stage		Use stage							End-of-life stage				
			A2 Transport A2 Transport A3 manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	D Reuse, recovery, recycling
Hazardous waste disposed (kg/FU)			3.46E-02	0.00E+00	6.92E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	7.17E-06	0.00E+00	0.00E+00	-4.58E-07
Non-hazardous waste disposed (kg/FU)			4.06E-01	2.25E-01	1.25E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	7.44E-02	0.00E+00	5.23E+00	-3.25E-03
Radioactive waste disposed (kg/FU)			2.53E-04	5.38E-05	7.25E-06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	3.80E-07	0.00E+00	4.68E-06	-4.55E-06
Components for re-use (kg/FU)			0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Materials for recycling (kg/FU)			4.12E-01	0.00E+00	2.69E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Materials for energy recovery (kg/FU)			0.00E+00	0.00E+00	1.44E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	0.00E+00	2.63E+00	0.00E+00	0.00E+00
Exported energy (MJ/FU)			0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.80E-01

9 IMPACT CATEGORIES ADDITIONAL TO EN 15804

		Production		Construction process stage		Use stage							End-of-life stage				
			A2 Transport A2 Transport A3 manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	D Reuse, recovery, recycling
	PM (disease incidence)		4.25E-07	2.40E-08	9.42E-09	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	5.43E-09	0.00E+00	3.93E-09	-9.55E-10
	IRHH (kg U235 eq/FU)		4.33E-01	2.49E-02	9.98E-03	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	5.86E-04	0.00E+00	2.69E-03	-8.16E-03
	ETF (CTUe/FU)		5.46E+01	4.80E+00	1.06E+00	1.51E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	6.02E-01	0.00E+00	4.38E-01	-3.13E-01
	HTCE (CTUh/FU)		5.96E-09	1.49E-10	1.27E-10	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	3.55E-11	0.00E+00	1.19E-11	-1.15E-11
	HTnCE (CTUh/FU)		2.39E-08	5.06E-09	7.90E-10	1.37E-11	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	1.05E-09	0.00E+00	3.70E-10	-1.96E-10
	Land Use Related impacts (dimensionless)		8.25E+01	3.39E+00	8.60E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	MND	MND	0.00E+00	8.81E-01	0.00E+00	1.03E+00	-1.28E+01

HTCE = HUMAN TOXICITY – CANCER EFFECTS; HTNCE = HUMAN TOXICITY – NON CANCER EFFECTS; ETF = ECOTOXICITY – FRESHWATER; (POTENTIAL COMPARATIVE TOXIC UNIT)

PM = PARTICULATE MATTER (POTENTIAL INCIDENCE OF DISEASE DUE TO PM EMISSIONS);

IRHH = IONIZING RADIATION – HUMAN HEALTH EFFECTS (POTENTIAL HUMAN EXPOSURE EFFICIENCY RELATIVE TO U235);

9.1 Environmental impact categories explained

	Global Warming Potential	The global warming potential of a gas refers to the total contribution to global warming resulting from the emission of one unit of that gas relative to one unit of the reference gas, carbon dioxide, which is assigned a value of 1. It is split up in 4: – Global Warming Potential total (GWP-total) which is the sum of GWP-fossil, GWP-biogenic and GWP-luluc – Global Warming Potential fossil fuels (GWP-fossil) : The global warming potential related to greenhouse gas (GHG) emissions to any media originating from the oxidation and/or reduction of fossil fuels by means of their transformation or degradation (e.g. combustion, digestion, landfilling, etc). – Global Warming Potential biogenic (GWP-biogenic) : The global warming potential related to carbon emissions to air (CO₂, CO and CH₄) originating from the oxidation and/or reduction of aboveground biomass by means of its transformation or degradation (e.g. combustion, digestion, composting, landfilling) and CO₂ uptake from the atmosphere through photosynthesis during biomass growth - i.e. corresponding to the carbon content of products, biofuels or above ground plant residues such as litter and dead wood. – Global Warming Potential land use and land use change (GWP-luluc): The global warming potential related to carbon uptakes and emissions (CO₂, CO and CH₄) originating from carbon stock changes caused by land use change and land use. This sub-category includes biogenic carbon exchanges from deforestation, road construction or other soil activities (including soil carbon emissions).
	Ozone Depletion	Destruction of the stratospheric ozone layer which shields the earth from ultraviolet radiation harmful to life. This destruction of ozone is caused by the breakdown of certain chlorine and/or bromine containing compounds (chlorofluorocarbons or halons), Which break down when they reach the stratosphere and then catalytically destroy ozone molecules.
	Acidification potential	Acid depositions have negative impacts on natural ecosystems and the man-made environment incl. buildings. The main sources for emissions of acidifying substances are agriculture and fossil fuel combustion used for electricity production, heating and transport.
	Eutrophication potential	The potential to cause over-fertilization of water and soil, which can result in increased growth of biomass and following adverse effects. It is split up in 3: – Eutrophication potential - freshwater: The potential to cause over-fertilization of freshwater, which can result in increased growth of biomass and following adverse effects. – Eutrophication potential - marine: The potential to cause over-fertilization of marine water, which can result in increased growth of biomass and following adverse effects. – Eutrophication potential - terrestrial: The potential to cause over-fertilization of soil, which can result in increased growth of biomass and following adverse effects.
	Photochemical ozone creation	Chemical reactions brought about by the light energy of the sun creating photochemical smog. The reaction of nitrogen oxides with hydrocarbons in the presence of sunlight to form ozone is an example of a photochemical reaction.
	Abiotic depletion potential for non-fossil resources	Consumption of non-renewable resources, thereby lowering their availability for future generations. Expressed in comparison to Antimony (Sb). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Abiotic depletion potential for fossil resources	Measure for the depletion of fossil fuels such as oil, natural gas, and coal. The stock of the fossil fuels is formed by the total amount of fossil fuels, expressed in Megajoules (MJ). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.



	Ecotoxicity for aquatic fresh water	The impacts of chemical substances on ecosystems (freshwater). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Human toxicity (carcinogenic effects)	The impacts of chemical substances on human health via three parts of the environment: air, soil and water. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Human toxicity (non-carcinogenic effects)	The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Particulate matter	Accounts for the adverse health effects on human health caused by emissions of Particulate Matter (PM) and its precursors (NO_x, SO_x, NH₃)
	Resource depletion (water)	Accounts for water use related to local scarcity of water as freshwater is a scarce resource in some regions, while in others it is not. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
	Ionizing radiation - human health effects	This impact category deals mainly with the eventual impact on human health of low dose ionizing radiation of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator.
	Land use related impacts	The indicator is the "soil quality index" which is the result of an aggregation of following four aspects: – Biotic production – Erosion resistance – Mechanical filtration – Groundwater The aggregation is done based on a JRC model. The four aspects are quantified through the LANCA model for land use. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.



10 DETAILS OF THE UNDERLYING SCENARIOS USED TO CALCULATE THE IMPACTS

10.1 A1 - raw material supply

Included. This module considers the extraction and processing of all raw materials and energy which occur upstream to the studied manufacturing process. Raw materials are modelled using primary process data for the stone wool. Literature data for the other components. Recycled content consists of secondary materials from the market and internally recycled materials. Both are modelled without environmental burden, apart from the processing it takes to prepare for on-site recycling.

10.2 A2 – transport to the manufacturer

Included. The materials are transported to the manufacturing site using supplier specific distances.

10.3 manufacturing

Included. This module considers the production process. The system boundaries include all materials, products and energy, as well as waste processing up to the end-of-waste state or disposal of final residues during the product stage. There are no co-products that have a value of 1% or higher and hence only mineral wool products and wastes leave the facility. In the case of input of secondary materials or energy recovered from secondary fuels, the system boundary between the system under study and the previous system (providing the secondary materials) is set where outputs of the previous system, e.g. materials, products or energy or building elements, reach the end-of-waste stage. No offsetting of CO₂ with certificates has been applied. Losses for manufacturing at ROCKWOOL are included.

10.4 A4 – transport to the building site

Fuel type and consumption of vehicle or vehicle type used for transport	Truck: Diesel, 1 litre per 3 km, by trailer
Distance	171 km
Capacity utilisation (including empty returns)	Truck: 85% delivery; 0% for 30% of empty returns
Bulk density of transported products	Varies from 34 kg/m ³ to 105 kg/m ³
Volume capacity utilisation factor	1 or lower for compressed insulation products

The products from the Roermond location for the Belgian market are shipped to customers (40%) using 150 km distance to Brussels and through a warehouse (60%) using the same distance plus 35 km for supply to the construction site. The 40:60 ratio has been checked and is representative for ROCKWOOL. The average distance is 171 km. Transportation of the insulation is by truck and is based on volume transportation. Contracted parties confirm a use of 1 litre of diesel per 3 km and a return freight that includes a 25% empty return with the rest being contracted out. The volume depends on the trailer type. ROCKWOOL operates normal trailers (88 m³), mega trailers (95 m³) and combi (105 m³) and load factors are typically high to optimize logistics and reduce cost. For this LCA a relatively small trailer is modelled based on a volume of 88 m³ with a load factor of 85%.



10.5 A5 – installation in the building

At the construction site, packaging materials are released. Also 2% material losses have been taken into account.

Parts of the installation	Quantity	Description
Processes necessary for the installation of the product		MANUAL INSTALLATION, SO NO ENERGY CONSUMPTION
Fixation materials		NOT DECLARED
Joining materials		NONE
Treatments		NONE
Material losses		2%
Packaging	0.002 0.005	KG PALLETS KG FOIL
Others		

Ancillary materials for installation (specified by material)	Insert information		
Water use	None		
Other resource use	None		
Quantitative description of energy type (regional mix) and consumption during the installation process	Manual installation		
Waste materials on the building site before waste processing, generated by the product's installation (specified by type)	No ancillary materials declared		
Output materials (specified by type) as result of waste processing at the building site e.g. of collection for recycling, for energy recovery, disposal (specified by route)	Not relevant		
Direct emissions to ambient air, soil and water	None		
Distance	Not relevant		



10.6 B – use stage (excluding potential savings)

Included. There are no consumables, maintenance, repair, replacements, or refurbishments related to the use of the ROCKWOOL insulation products.

Operational benefits from using insulation material on the building level have not been included as the building performance is unclear without a specific building design.

Emissions during use are included.

B1 The free formaldehyde content in the product is the one that could be potentially released through the lifetime of the product and therefore this approach serves as an indication to the figure assigned to module B1. The default value will therefore be 16 mg/kg.

ROCKWOOL insulation products for the French market have been tested and obtained a category A+ (très faibles émissions, or very low emissions) rating based on the French “Plan National Santé Environnement (PNSE)”.

B2 ROCKWOOL products do not require maintenance during use in standard conditions and if correctly applied.

B3 ROCKWOOL building insulation products are hidden from the surface and do not require any cleaning.

B4 ROCKWOOL products will not be replaced during use in standard conditions and if correctly applied; (correctly applied meaning that installed according to ROCKWOOL instructions).

B5 ROCKWOOL products are not refurbished during use in standard conditions and if correctly applied.

B6 and B7 relate to the operations of the building and have therefor not been declared.

10.7 C: End of life

The deconstruction is manual like the installation. The transportation distances and waste processing and landfill scenarios are identical to the building waste scenario.

Materials that are landfilled include manual dismantling in module [C1] and transportation in [C2] and emissions from landfilling and incineration in [C4].



Module C2 – Transport to waste processing						End-of-life modules – C3 and C4	
Type of vehicle (truck/boat/etc.)	Fuel consumption (litres/km)	Distance (km)	Capacity utilisation (%)	Density of products (kg/m3)	Assumptions	Parameter	Value (kg)
Truck	0.37	30+100	50	35	incineration	Wastes collected separately	0
						Wastes collected as mixed construction waste	100
						Waste for re-use	0
						Waste for recycling	0
						Waste for energy recovery	50
Truck	0.37	30+50	50	35	landfill	Waste for final disposal	50

10.8 D – Benefits and loads beyond the system boundaries

Information module D aims at transparency for the environmental benefits or loads resulting from reusable products, recyclable materials and/or useful energy carriers leaving a product system, e.g. as secondary materials or fuels. Any declared benefits and loads from net flows leaving the product system that have not been allocated as co-products and that have passed the end-of-waste state are included in module D. This includes the waste from [A3] that is recycled and the packaging from [A5]. There are no benefits associated with the ROCKWOOL products themselves declared under module D.

QUANTITATIVE DESCRIPTION OF THE LOADS BEYOND THE SYSTEM BOUNDARIES	Wood that is burned is treated as thermal waste treatment and not energy recovery as the efficiency is less than 60%. The heat recovery is modelled as part of module [D]. When wood is combusted, the renewable energy is consumed, part of the energy is recovered, and the biogenic carbon is released as carbon dioxide. The recovery is modelled using the lower heating value of 13.99 MJ/kg and a production of heat of 20% and electricity of 10% using the process for heat production: "Thermal energy from natural gas" (e.g. Heat, natural gas, at industrial furnace > 100kW) and for electricity production: Belgian high voltage consumption mix, excluding the impact from the transmission network.
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11 RELEASE OF DANGEROUS SUBSTANCES TO INDOOR AIR, SOIL AND WATER DURING THE USE STAGE

11.1 Indoor air

The European Regulation (CE) n° 1272/2008 on classification, labelling and packaging of substances and mixtures updated by the Regulation (CE) n° 790/2009 does not classify mineral wool fibres as a dangerous substance as long as they fulfil criteria defined in its Note Q. The European Certification Board for mineral wool products, EUCB, certify mineral wool products made of fibres are fulfilling Note Q. www.euceb.org. The product subject of this EPD has a certificate demonstrating compliance to Note Q. According to IARC, mineral wool insulating products are not classifiable as to carcinogenicity in humans (group 3). The fibers are only to be expected to be released when the product is handled, during installation. Once installed there are is no release of fibres.

ROCKWOOL insulation products for the French market have been tested and obtained a category A+ (très faibles émissions, or very low emissions). The product subject of this EPD emits formaldehyde in concentrations < 0.0016% which is way below the CLP threshold of 0,1%.

11.2 Soil and water

There are no emissions to soil and water during the use stage.



12 DEMONSTRATION OF VERIFICATION

EN 15804+A2 serves as the core PCR.

Independent verification of the environmental declaration and data according to standard EN ISO 14025:2010.

Internal ☐ External ☒

Third party verifier: Agnes Schuurmans, SGS INTRON, Dr. Nolenslaan 126, 6136 GV Sittard, agnes.schuurmans@sgs.com



13 APPLICATION UNIT

PRODUCT	DESCRIPTION	REFERENCE FLOW	THICKNESS (MM)	SCALABILITY MINIMUM (MM)	SCALABILITY MAXIMUM (MM)	SPECIFICATIONS	APPLICATION	RATIO	
Rockfit Mono	mineral wool, 600 x 1000 mm, λ 0.035 W/m.K	1 m ²	150	40	250	Euro Class A1; unit load 35 kg/m ³ , R _p =4.25; no facing; loss of ignition = 3.2	Cavity walls; curtain walls; separating walls	1 m ²	1



14 ADDITIONAL INFORMATION ON REVERSIBILITY

Description	Type of fixing	Level of reversibility	Simplicity of disassembly	Speed of disassembly	Ease of handling (size and weight)	Robustness of material (material resistance to disassembly)	Comment
Insulation of masonry cavity walls with partial or full cavity filling	Fixed with cavity ties and clips (components to be added)	Reversible connections with non-repairable damage	Simple – no specific dismantling tools required	Speedy disassembly	Easy to manipulate (by hand (small size and limited weight): one worker should be sufficient	Disassembly is possible but should be done carefully in order not to generate any damage	

15 BIBLIOGRAPHY

- ISO 14040:2006: Environmental Management-Life Cycle Assessment-Principles and framework.
- ISO 14044:2006: Environmental Management-Life Cycle Assessment-Requirements and guidelines.
- ISO 14025:2006: Environmental labels and Declarations-Type III Environmental Declarations-Principles and procedures.
- NBN EN 15804+A2:2019
- B-EPD-PCR
- EN13162+A1:2015 Thermal insulation products for buildings. Factory made mineral wool (MW) products. Specification, 4.2.1
- EN12667:2000 Thermal performance of building materials and products - Determination of thermal resistance by means of guarded hot plate and heat flow meter methods - Products of high and medium thermal resistance.
- EN12939:2001 Thermal performance of building materials and products. Determination of thermal resistance by means of guarded hot plate and heat flow meter methods. Thick products of high and medium thermal resistance."
- EN13162+A1:2015 Thermal insulation products for buildings. Factory made mineral wool (MW) products. Specification, 4.3.15



General information

Owner of the EPD, Responsible for the data, LCA and information



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Verifier



Agnes Schurmans, SGS INTRON
Date of verification: 24.10.2023
External independent verification of the declaration and data
according to EN ISO 14025 and relevant PCR documents

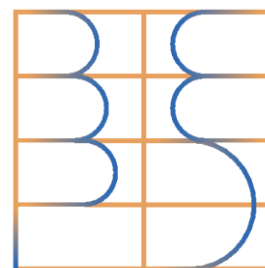
Comparing EPDs is not possible unless they are conform to the same PCR and taking into account the building context.
The program operator cannot be held responsible for the information supplied by the owner of the EPD nor LCA practitioner.



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